AML-3406

AL and ML Capstone Project

# Capstone Project Weekly Progress Report

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| **Semester** | Fall-2024 Semester 3rd |
| **Course Code** | AML-3406 |
| **Section** | Section 1 |
| **Project Title** | Car Damage Detection |
| **Group Name** | Group B |
| **Student names/Student IDs** | Sakshi (C0908000) – TL<br>Bansil Patel (C0912873)<br>Harsh Mohile (C0912872)<br>Meet Patel (C0910378)<br>Rachit Bhatt (C0902810) |
| **Reporting Week** | Week 12 |
| **Faculty Supervisor** | William Pourmajidi |

**1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)**

| Sakshi | Bansil Patel | Harsh Mohile | Meet Patel | Rachit Bhatt |
|---|---|---|---|---|
| Good | Good | Good | Good | Good |

## 2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

### 2.1 Individual tasks (each team member)

**Sakshi:** Enhance the GitHub Actions pipeline by adding real-time monitoring and alerting features to track deployment status.

**Bansil Patel & Meet Patel:** Plan to expand testing with more diverse datasets, focusing on rare car damage types for improved model generalization**.**

**Harsh Mohile & Rachit Bhatt:** Further optimize the Deployment UI to improve user experience and make it more intuitive for end users.

### 2.2 Your team's tasks

- Model Enhancements: Plan to explore advanced model architectures to improve accuracy, aiming for higher performance in detecting complex car damages.
- Deployment Pipeline: Add real-time monitoring to the deployment pipeline, enabling better tracking of deployment success and quick rollback if needed.
- UI Optimization: Improve the Deployment UI with new features such as interactive visualizations of model predictions and user feedback integration.
- Scalability: Focus on optimizing the model's performance for large-scale data processing to handle a wider variety of car images in real-world applications.

## 3. Progress made in Reporting Week at an individual level and group level

### 3.1 Individual progress

**Sakshi:** Resolved repository configuration issues to ensure consistent deployments.

**Bansil Patel & Meet Patel:** Validated the model's robustness by testing on multiple datasets, confirming its generalization to different car damage types.

**Harsh Mohile & Rachit Bhatt:** Conducted extensive testing across staging and production environments, ensuring reliable performance.

### 3.2 Your team's progress

- Finalized the automated deployment pipeline with GitHub Actions, ensuring smooth deployments across all environments. Fully integrated the UI with the backend, ensuring seamless deployment in staging and production.

## 4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

### 4.1 Individual project blockers

**Sakshi:** Minor integration issues were resolved by refining repository configurations.
**Bansil Patel & Meet Patel:** Delays in backend-UI communication but resolved during testing.
**Harsh Mohile & Rachit Bhatt**: N/A.

### 4.2Your team's project blockers

- Multi-branch setup caused delays, resolved with configuration adjustments.
- Minor adjustments to align the UI with production configurations.

## 5. Tasks to be completed in the next week at the individual level and group level

### 5.1 Individual tasks

**Sakshi, Bansil Patel & Meet Patel:** N/A

**Harsh Mohile & Rachit Bhatt:** N/A

### 5.2 Your team's tasks

The project has been completed successfully.

**6. Include the tasks from your sprint planning (GitHub/ZenHub) for the present period.**

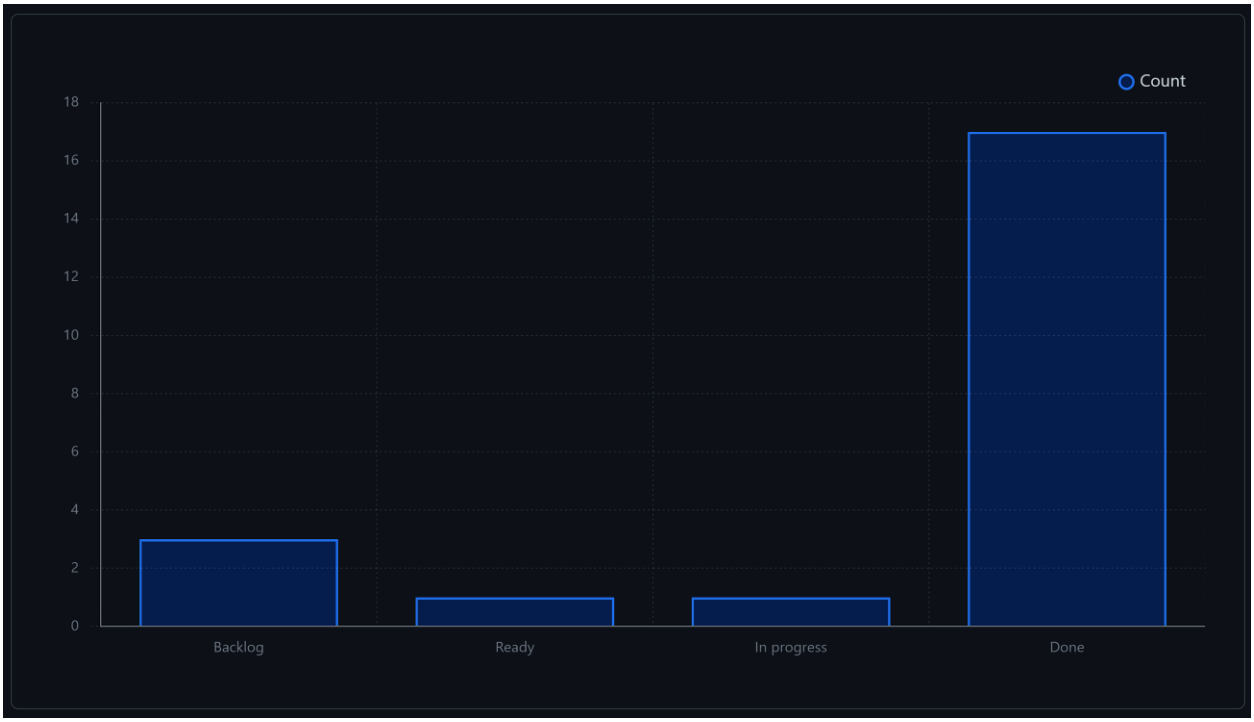


**Figure 1:** Weekly Activities

- In this week, there were a lot of issues resolved from the backlog, contributing to development.

**7. Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.**

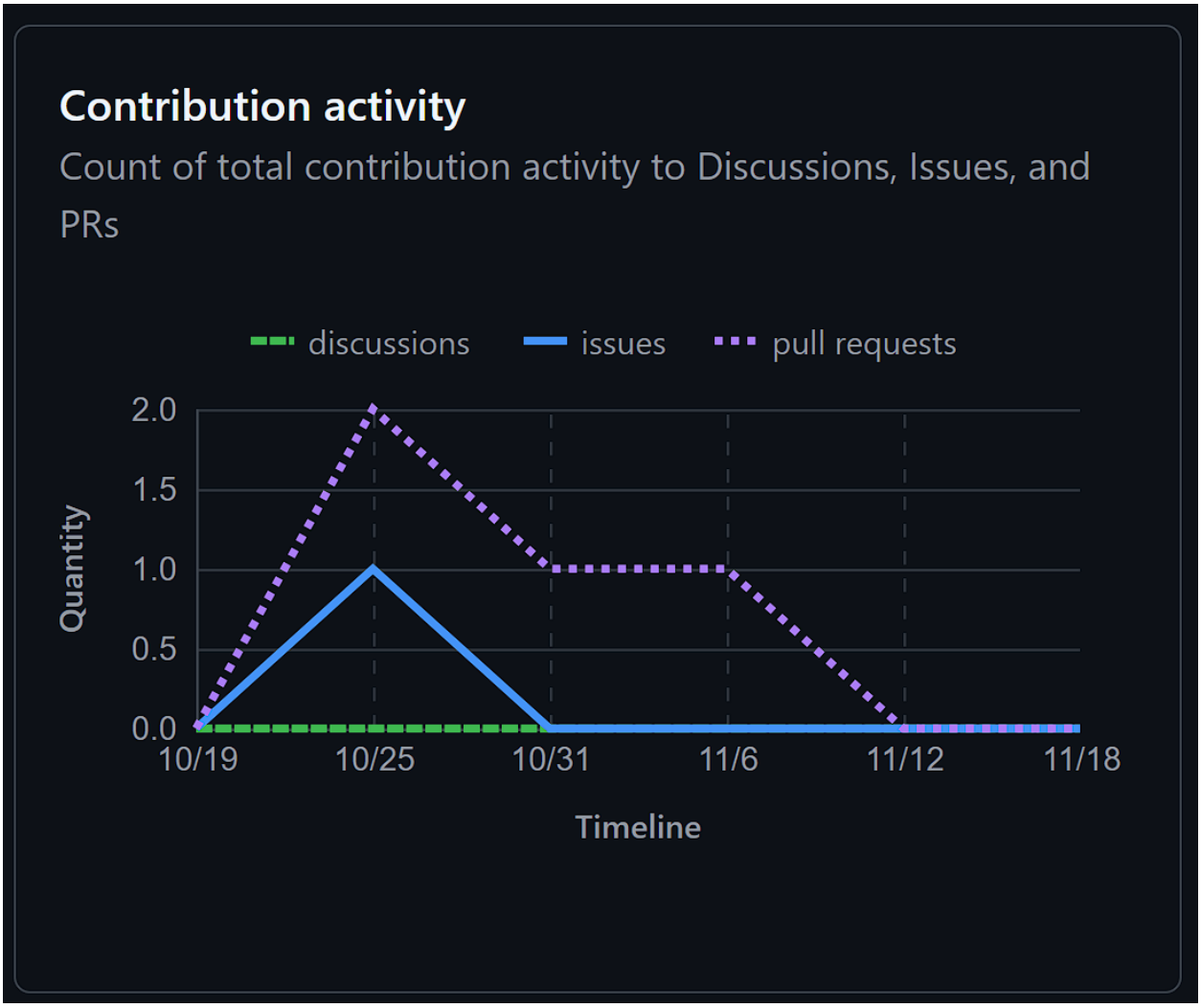


**Figure 2:** Contributors in the Week

- This week, we had a few PRs and issues resolved as compared to previous week, demonstrating team hard work and dedication.

8. **Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.**

**GitHub:** https://www.github.com/rachit-bhatt/Capstone-Project